How Restricting Migrants' Job Options Affects Both Migrants and Residents

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These results are not official statistics. They have been created for research purposes from the Integrated Data Infrastructure (IDI) which is carefully managed by Stats NZ. For more information about the IDI please visit `https://www.stats.govt.nz/integrated-data/`. The results are based in part on tax data supplied by Inland Revenue to Stats NZ under the Tax Administration Act 1994 for statistical purposes. Any discussion of data limitations or weaknesses is in the context of using the IDI for statistical purposes, and is not related to the data's ability to support Inland Revenue's core operational requirements. Access to the data used in this study was provided by Stats NZ under conditions designed to give effect to the security and confidentiality provisions of the Data and Statistics Act 2022. The results presented in this study are the work of the author, not Stats NZ or individual data suppliers.

Can work visas facilitate immigration while protecting residents' wages?

We focus on one aspect of work visa policy: **Requirements that migrants work in certain jobs.**

- 'High-skill' occupations.
- Rural areas or certain cities.
- The agriculture or medical sectors.
- Particular firms.

We ask how these requirements affect **migrant workers**, **resident workers**, and their **employers**.

Specifically: We study New Zealand's Essential Skills work visa. These migrants were **limited to positions for which no New Zealanders were believed to be available**.

Incidence will depend on how firms set wages:

Bargaining?

- Firms bargain with individual workers.
- Weaker outside options decrease migrants' bargaining power and so decrease their wages.
- **Residents only affected insofar as their marginal product is affected.**

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- Firms commit to position-specific wages.
- A firm's wage choice will depend on the distribution of outside options.
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A fundamental topic in labor economics with implications for the **causes of wage inequality**, the **effects of employment regulations** and the **allocative efficiency** of the labor market (Flinn & Mullins '21; Silbert & Townsend '24).

Today's talk

Reduced-form facts:

- Migrants earn less than other workers — but only because of the jobs they hold.
- Loosening an *individual* migrant's job restrictions does not affect their wages.
- But loosening the restrictions on *all* migrants within an occupation does increase wages.

These facts are consistent with a model in which wages are set through monopsonistic wage-posting.

We estimate such a model:

- Restricting migrants' job options decreases their wages by an average of 6%.
- Most residents are unaffected — but 2.5% have their wage decreased by $\geq 2\%$.
- The restrictions decrease aggregate welfare — mostly because of non-wage amenities.

Existing literature: work visas

- Individual workers' visa status matters little (Wang '21),
- While the overall migrant visa system matters a lot (Naidu, Nyarko & Wang '16; Ahrens, et. al '23; Borjas & Edo '23).

We reconcile these results by **combining analyses of individual- and market-level shocks**.

We also extend this literature by asking how **visa systems affect residents' wages** (Amior & Manning '20; Amior & Stuhler '23; Borjas & Edo '23).

Existing literature: theories of wage-setting

- How workers' outside job options affect their wages (Caldwell & Harmon '19; Caldwell & Danieli '22).
- Other predictions (Carvalho, da Fonseca & Santarrosa '23; Bloesch, Larsen & Taska '22; Hall & Krueger '12; Roussille & Scuderi 24; Rubens & Delabastita '24).

We study the effect of outside options **randomized** at the **individual level**.

We identify the effect of **better outside options on workers who don't move jobs**.

The Essential Skills Work Visa

- Existed between 2008 and 2022.
- Accounted for 80% of employer-assisted visas, and about 25% of all work visas.
- Covers a broad range of skilled and semi-skilled occupations. Most common: chef, carpenter, dairy cattle farmer.

The Essential Skills visa was limited to positions for which there were 'no New Zealanders available'.

- For most occupations: must show a 'genuine attempt' to recruit New Zealanders
- ... but **occupations on an "Essential Skills in Demand list" are exempt**.

Data and sample

We use linked **census**, **tax** and **immigration** data:

- **Census data:** Employment status, hours, occupation, education and demographics, in 2013 and 2018.
- **Tax data:** Worker-month-firm panel of earnings & location.
- **Immigration data:** Spell data on visas held, border movements, and occupations. Entries into resident visa lotteries.

Between **August 2008 to March 2022**: 278m observations across **3.8m workers**.

Among **205k Essential Skills workers**:

- 32% live in Auckland (vs. 27% overall). 54% urban (vs. 51% overall).
- Average age 35. 71% are men.
- 38% have at least a degree (vs. 32% overall).

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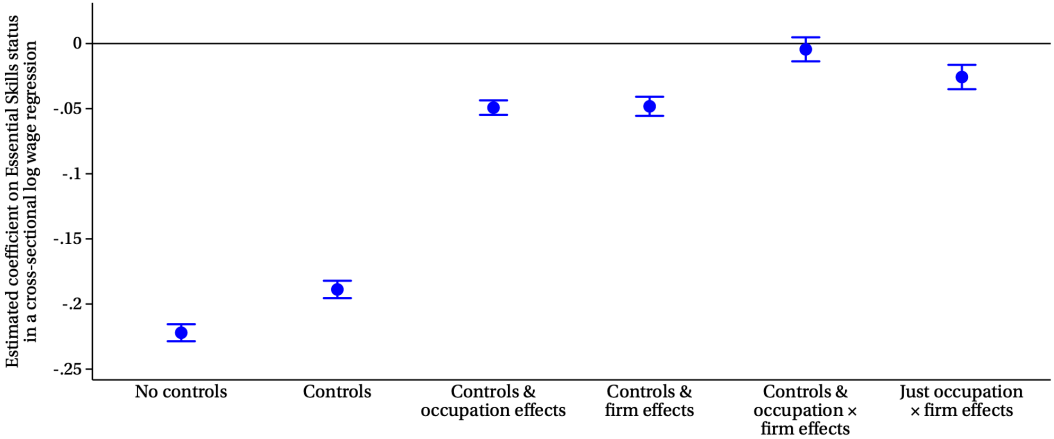
Migrants earn less than other workers...

Sample: full-time employees with a unique firm in the 12 months before the 2018 Census.

Migrants earn less than other workers... but only because of their jobs



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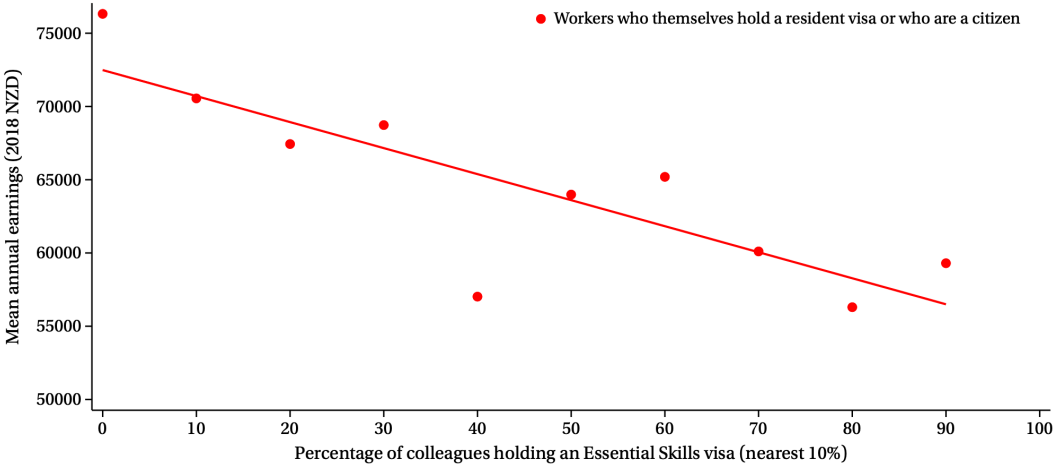

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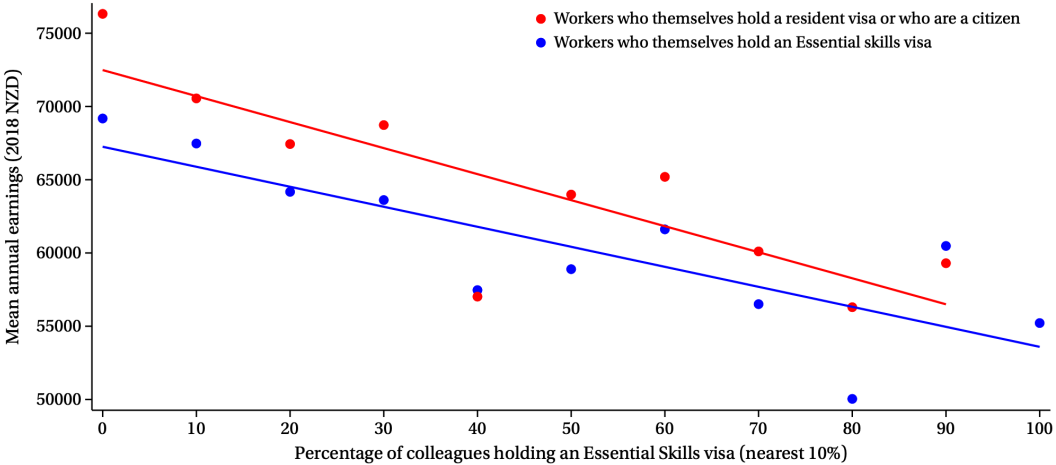

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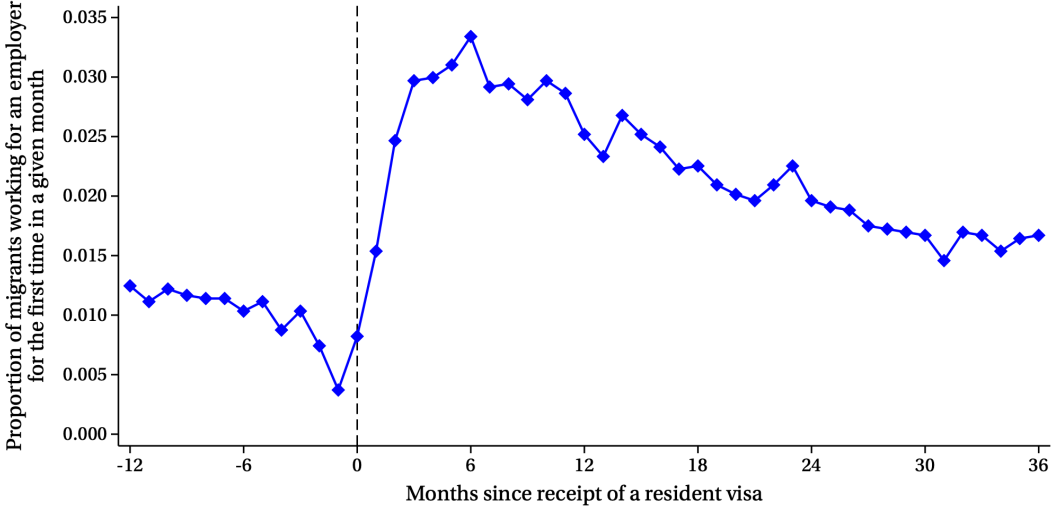

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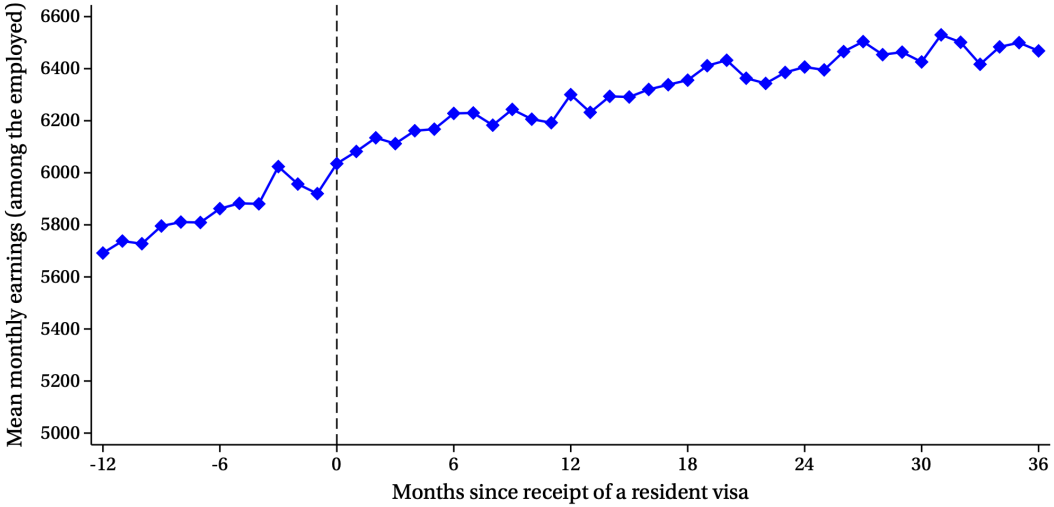

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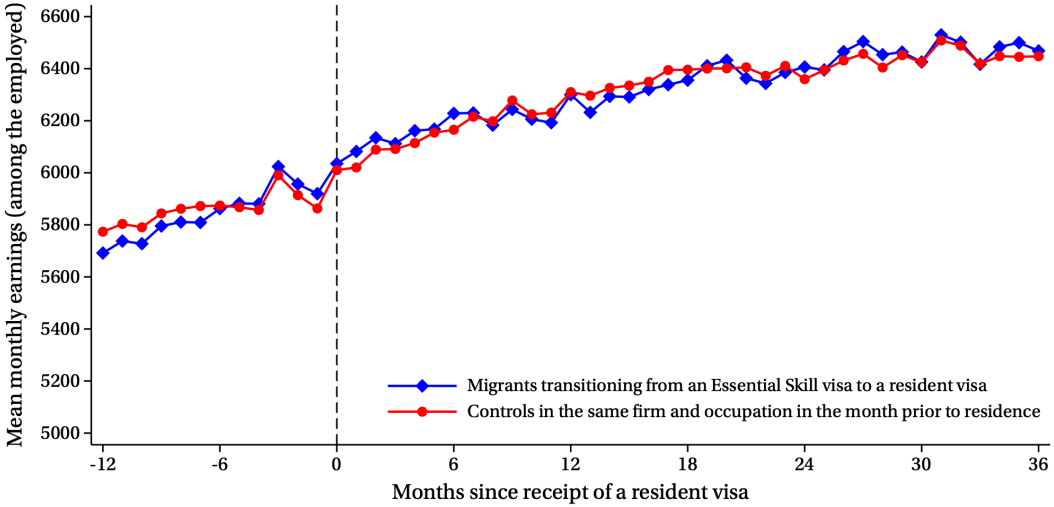
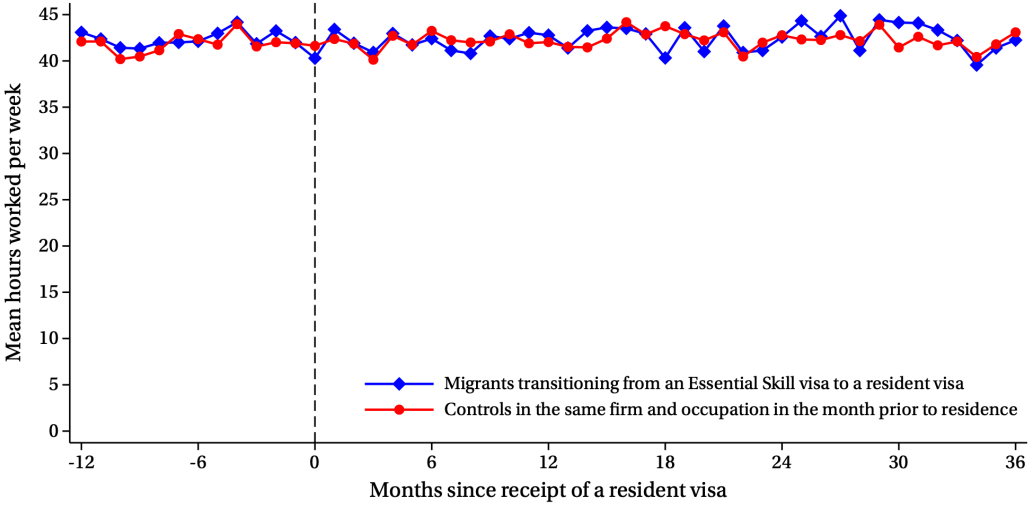


An individual migrant's job restriction does not affect their wage

Concern 1: New residents are able to negotiate better working conditions.

- Solution: *Study self-reported hours of work* in labor force surveys.

Obtaining a resident visa has no effect on hours of work



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In an occupation, strengthening *all* migrants' job options increases wages

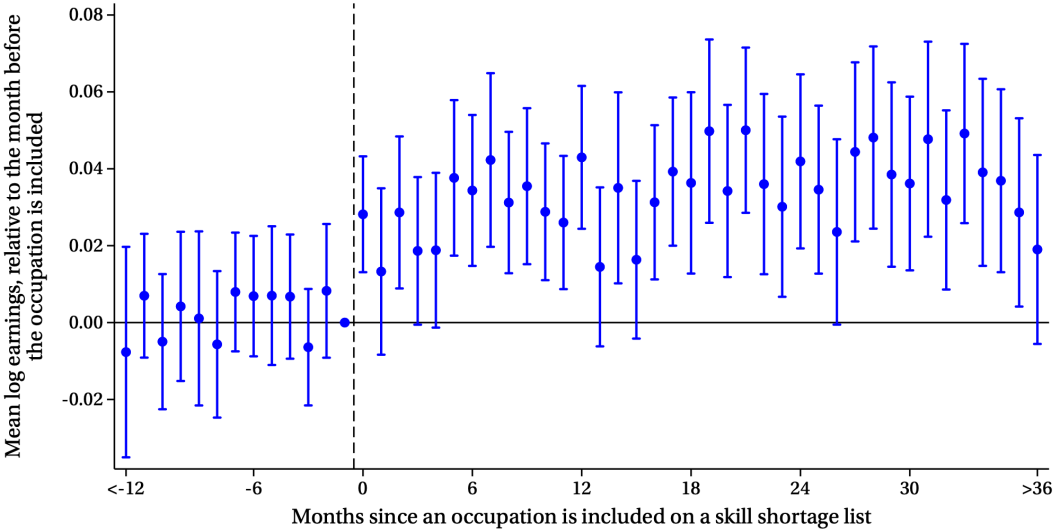
For **occupations on an 'Essential Skills in Demand' list**:

- Prospective employers need not show a 'genuine attempt' to recruit New Zealanders.
- So, **Essential Skills migrants' job options are unrestricted**.

Occupations were added and removed from these lists regularly, often for reasons orthogonal to demand (*e.g.* occupation size).

We ask how wages change when an occupation is listed.

In an occupation, strengthening *all* migrants' job options increases wages



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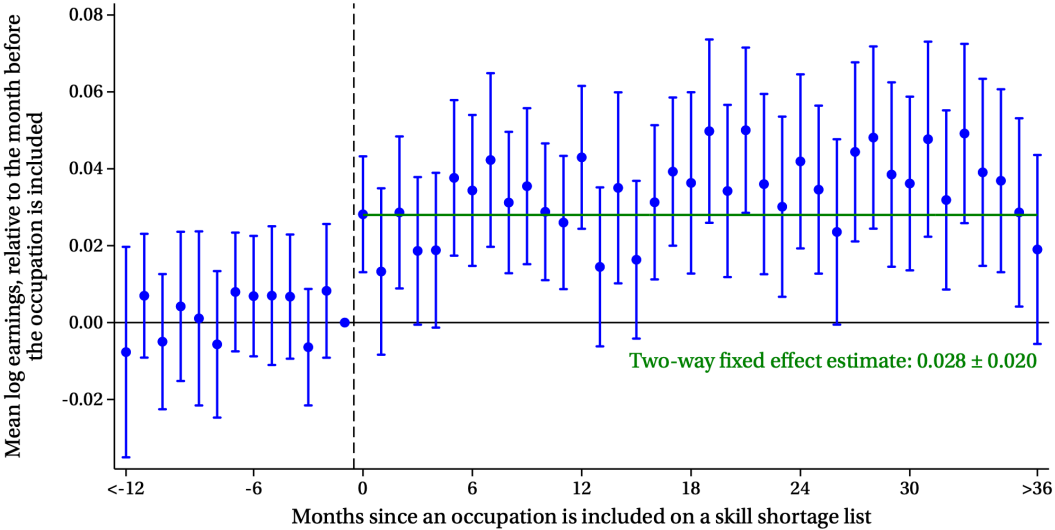

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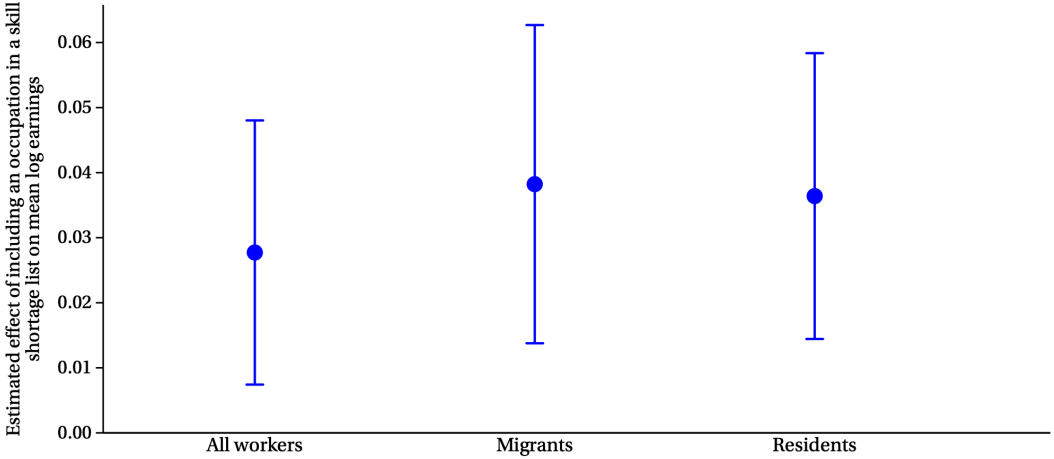
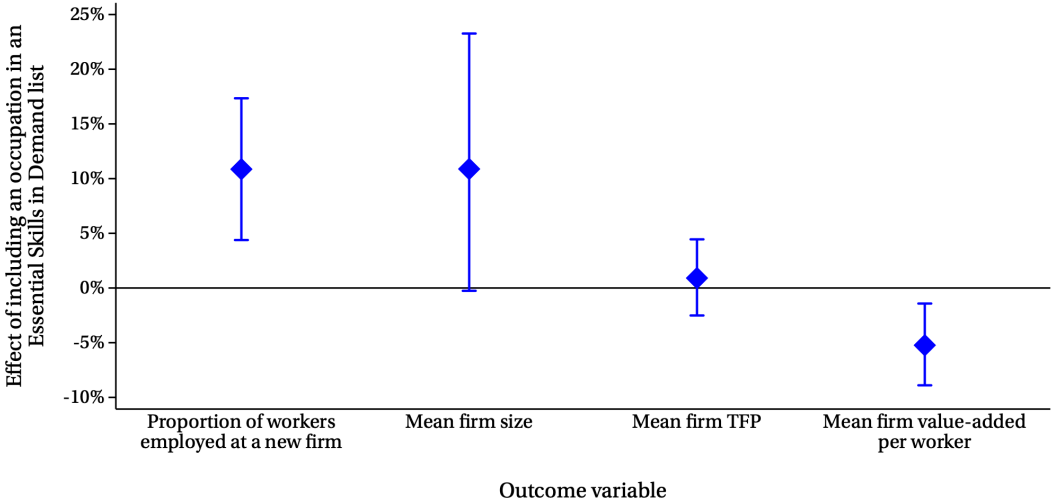
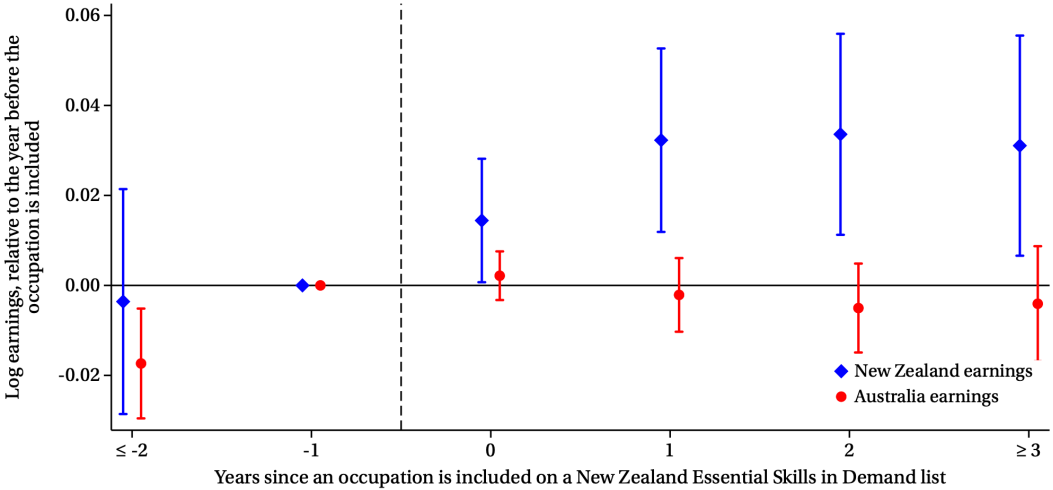


Why does strengthening migrants' job options increase wages?



Using Australian data as a placebo outcome



Point estimate.

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Structural model: motivation

Research question: **how does restricting migrants job options affect wages, profits, and welfare?**

Limitation of reduced-form analyses:

- Can't identify the distribution of effects across workers.
- Firm-level data is noisy and often missing: can't estimate profit effects.
- Don't observe non-wage amenities: can't identify welfare effects.

Structural model: overview

The model is designed to capture two mechanisms.

Restricting migrants' job options will **decrease the elasticity of their labor supply, inducing firms to set less generous wages**.

- Empirical question: how restricted are migrants' job options in practice?
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Restricting migrants' job options will **segregate them into certain firms, decreasing their marginal product**.

- Empirical question: How concave is production? How easily can firms substitute across occupations?
- *Estimate production functions which account for diseconomies of scale and substitution across occupations.*

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Firms:

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⇒ **Firms which employ many migrants will face less elastic labor supply**.
⇒ **These firms will pay a lower wage — to both migrants and non-migrants.**

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Key assumption: shocks to the amenity-value of employment are orthogonal to productivity shocks.

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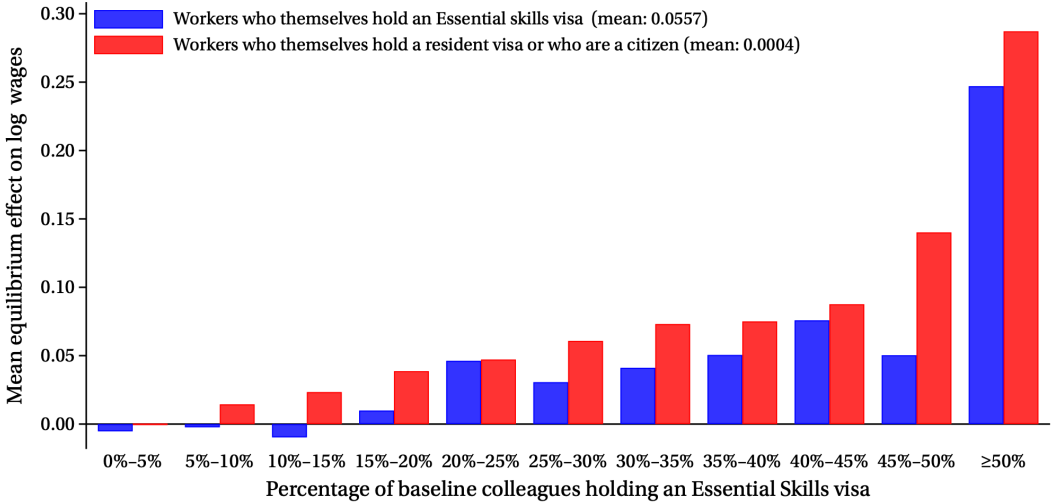
Solution: calculate wages in each of two random split-samples. Require productivity shocks in one sample be orthogonal to amenity shocks in the other sample.

Structural model: estimates

	Homogeneous coefficients	Heterogeneous coefficients	
Labor supply estimates		NZ-born	foreign-born
τ (labor supply elasticity for an atomistic firm)	0.814	0.845	0.751
	[0.197]	[0.208]	[0.183]
λ (workers' willingness substitute across space)	0.131	0.118	0.153
	[0.008]	[0.008]	[0.010]
Production function estimates			
ρ (firms' ability to substitute across occupations)	0.919	0.919	
	[0.019]	[0.019]	
ν (returns to scale)	0.940	0.940	
	[0.014]	[0.014]	

The equilibrium effect of allowing migrants to work at any firm



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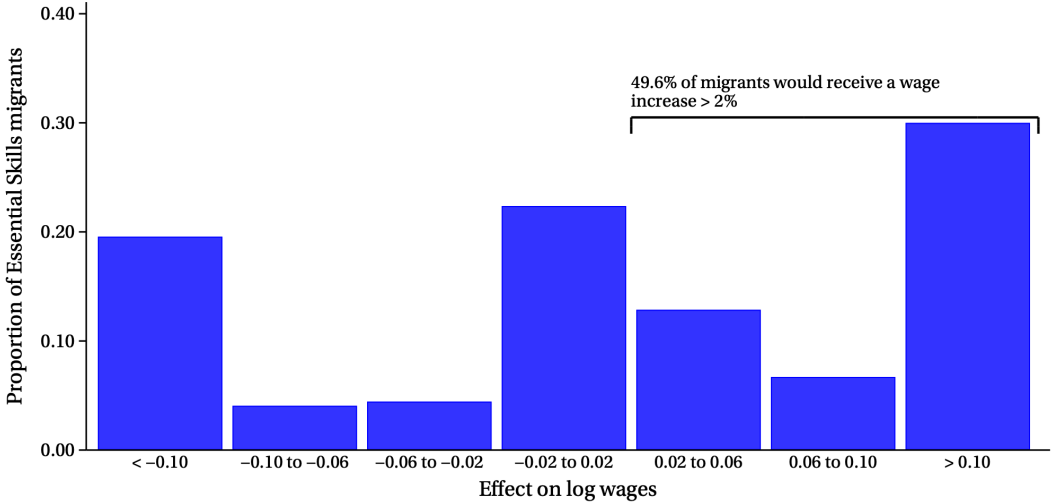

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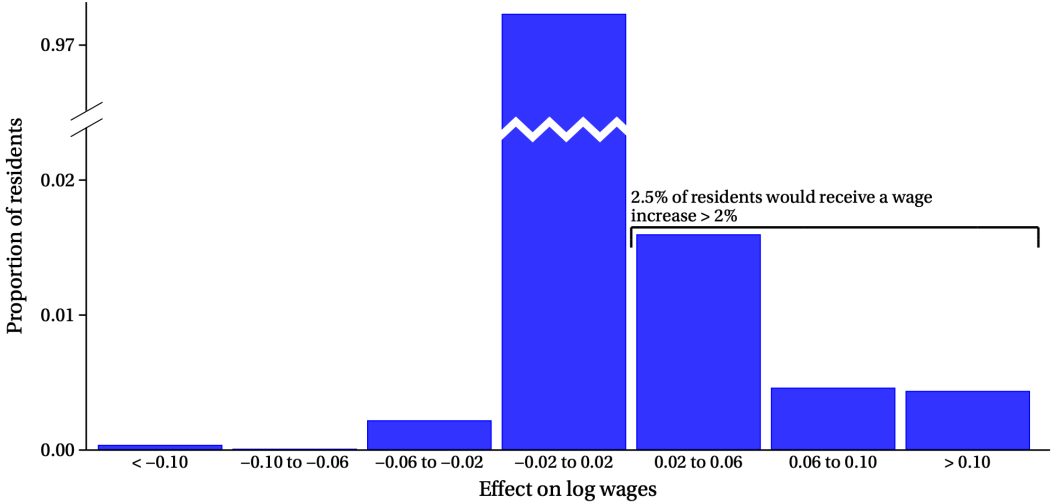

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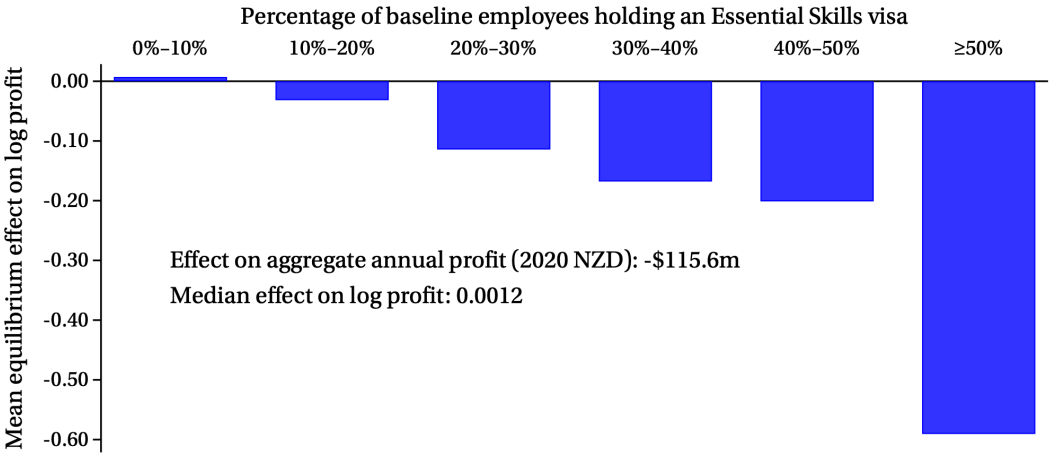

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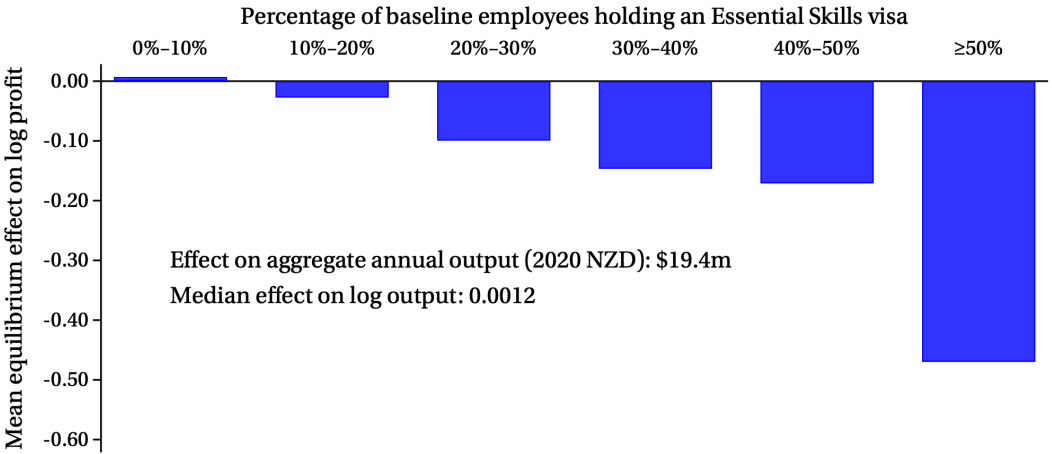

How restricting migrants' job options affects welfare

Let WTP_i denote i's money-metric welfare gain from the counterfactual equilibrium w', f', in which migrants can work at any firm:

$$u_i\left(w'_{o_i,f'_i} - WTP_i,\, f'_i\right) = u_i\left(w_{o_i,f_i},\, f_i\right),$$

which we can decompose into

- an earnings effect $w'_{o_i,f'_i} - w_{o_i,f_i}$, and
- a non-penuniary effect $WTP_i - \left(w'_{o_i,f'_i} - w_{o_i,f_i}\right)$.

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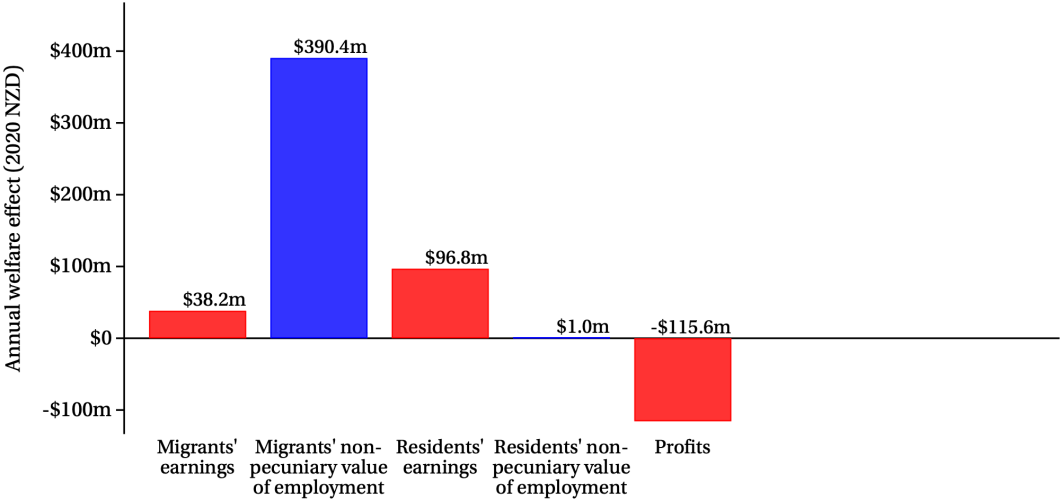

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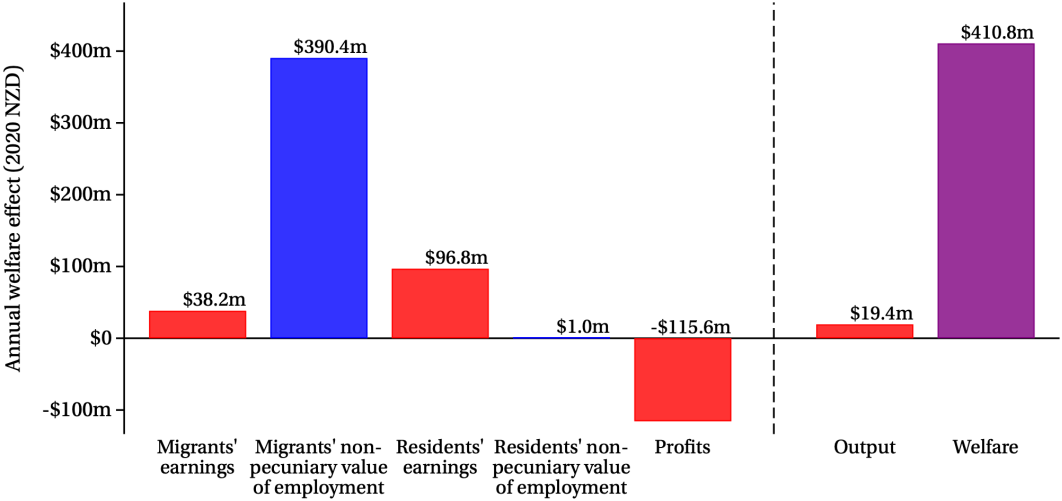

Restricting migrants' job options affects everyone's wages

Firms don't *specifically* discriminate against migrants,

But firms do account for migrants' weaker job options when setting everyone's wages.

Restricting migrant's job options reduces both their wages and the wages of their colleagues.